

Statistics

Question1

Variance of the following continuous frequency distribution is

Class Interval	Frequency
0 – 4	1
4 – 8	2
8 – 12	2
12 – 16	1

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Options:

A.

16

B.

$\frac{44}{3}$

C.

23

D.

$\frac{22}{3}$

Answer: B

Solution:

Class interval	Frequency (f_i)	x_i	$f_i x_i$	$f_i x_i^2$
0 – 4	1	2	2	4
4 – 8	2	6	12	72
8 – 12	2	10	20	200
12 – 16	1	14	14	196
$\sum f_i = N$ = 6		$\sum f_i x_i$ = 48	$\sum f_i x_i^2$ = 472	

So, mean ($\bar{x}$) = $\frac{\sum f_i x_i}{N} = \frac{48}{6} = 8$

and variance (σ^2) = $\frac{\sum f_i x_i^2}{N} - (\bar{x})^2$
 $= \frac{472}{6} - 8^2 = \frac{236}{3} - 64$
 $= \frac{236 - 192}{3} = 44/3$

Question2

The mean deviation from the median for the following data is

x_1	9	3	7	2	5
f_1	1	6	2	8	4

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Options:

A.

$$\frac{94}{21}$$

B.

$$\frac{12}{7}$$

C.

$$\frac{10}{7}$$

D.

$$\frac{100}{21}$$

Answer: C

Solution:

x_i	f_i	cf	$ x_i - 3 $	$f_i x_i - 3 $
2	8	8	1	8
3	6	14	0	0
5	4	18	2	8
7	2	20	4	8
9	1	21	6	6

$\Rightarrow N = 21 \Rightarrow \frac{N}{2} = 10.5$
 $\therefore \text{Median} = M = 3$
and Mean deviation about median
$$= \frac{\sum f_i |x_i - M|}{\sum f_i} = \frac{30}{21} = \frac{10}{7}$$

Question3

The mean deviation about the mean for the following data is

Class Interval	0 – 2	2 – 4	4 – 6	6 – 8	8 – 10
Frequency	1	3	4	1	2

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Options:

A.

3

B.

$$\frac{20}{11}$$

C.

$\frac{40}{11}$

D.

2

Answer: B

Solution:

Class interval	0-2	2-4	4-6	6-8	8-10	
Frequency	1	3	4	1	2	
CI	x_i	f_i	$x_i f_i$	$x_i - \bar{x}$	$f_i x_i - \bar{x} $	
0-2	1	1	1	-4	+4	
2-4	3	3	9	-2	+6	
4-6	5	4	20	0	0	
6-8	7	1	7	2	2	
8-10	9	2	18	4	8	

$$\sum f_i = 11, \sum f_i x_i = 55$$
$$\bar{X} = \frac{\sum f_i x_i}{\sum f_i} = \frac{55}{11} = 5$$
$$\sum f_i |x_i - \bar{x}| = 20$$

∴ Mean deviation about mean is

$$\frac{\sum f_i |x_i - \bar{x}|}{\sum f_i} = \frac{20}{11}$$

Question4

Let $x_1, x_2, \dots, x_{11}$ be the observations satisfying $\sum_{i=1}^{11} (x_i - 4) = 22$ and $\sum_{i=1}^{11} (x_i - 4)^2 = 154$. If the mean and variance of the

observations are α and β , then the quadratic equation having the roots $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ is

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Options:

A.

$$15x^2 - 16x + 15 = 0$$

B.

$$15x^2 - 34x + 15 = 0$$

C.

$$x^2 - 16x + 60 = 0$$

D.

$$12x^2 - 25x + 20 = 0$$

Answer: B

Solution:

$$\sum_{i=1}^{11} (x_i - 4) = 22$$

$$\Rightarrow \sum_{i=1}^{11} x_i - 4 \times 11 = 22$$

$$\Rightarrow \sum_{i=1}^{11} x_i = 66$$

So, mean $(\bar{x}) = \frac{66}{11} = 6$ and ... (i)

$$\sum_{i=1}^{11} (x_i - 4)^2 = 154$$

$$\begin{aligned} \therefore x_i - \bar{x} &= x_i - 6 \\ &= (x_i - 4) - 2 \\ \therefore [x_i - \bar{x}]^2 &= [(x_i - 4) - 2]^2 \\ &= (x_i - 4)^2 - 4(x_i - 4) + 4 \end{aligned}$$

$$\begin{aligned}
 \text{So, } \sum_{i=1}^{11} (x_i - \bar{x})^2 &= \sum_{i=1}^{11} (x_i - 4)^2 \\
 &\quad - 4 \sum_{i=1}^{11} (x_i - 4) + 11 \times 4 \\
 &= 154 - 4 \times 22 + 44 = 110
 \end{aligned}$$

$$\begin{aligned}
 \text{Variance } (\sigma) &= \frac{1}{n} \sum (x_i - \bar{x})^2 \\
 &= \frac{1}{11} \times 110 = 10
 \end{aligned}$$

Thus, $\alpha = 6, \beta = 10$

$$\text{Therefore, } \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{6}{10} + \frac{10}{6} = \frac{3}{5} + \frac{5}{3}$$

$$= \frac{9+25}{15} = 34/15 = \text{Sum of roots}$$

$$\text{and } \frac{\alpha}{\beta} \cdot \frac{\beta}{\alpha} = 1 = \text{product of roots}$$

Hence, the quadratic is

$$\begin{aligned}
 x^2 - \left(\frac{34}{15}\right)x + 1 &= 0 \\
 \Rightarrow 15x^2 - 34x + 15 &= 0
 \end{aligned}$$

Question 5

The mean and variance of the observations $x_1, x_2, x_3 \dots x_{15}$ are respectively 2 and 4 . If the mean and variance of the observations $y_1, y_2 \dots, y_{10}$ are respectively 2 and 5 , then the variance of the observations $x_1, x_2 \dots, x_{15}, y_1, y_2 \dots, y_{10}$ is

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Options:

A.

6.5

B.

5.3

C.

3.4

D.

4.4

Answer: D

Solution:

Use formula for combined variance

Let $n_1 = 15, \mu_1 = 2, \sigma_1^2 = 4$

$n_2 = 10, \mu_2 = 2, \sigma_2^2 = 5$

$$\begin{aligned}\text{Combined mean} = \mu &= \frac{n_1\mu_1 + n_2\mu_2}{n_1 + n_2} \\ &= \frac{15 \times 2 + 10 \times 2}{25} = 2\end{aligned}$$

Combined variance $= \sigma^2$

$$= \frac{n_1\sigma_1^2 + n_2\sigma_2^2 + n_1(\mu_1 - \mu)^2 + n_2(\mu_2 - \mu)^2}{n_1 + n_2}$$

Since, $\mu_1 = \mu_2 = \mu$ the last two terms becomes zero.

$$\begin{aligned}\text{So, } \sigma^2 &= \frac{15 \times 4 + 10 \times 5}{25} = \frac{60 + 50}{25} = \frac{110}{25} \\ &= 4.4\end{aligned}$$

Question6

Variance of the following discrete frequency distribution is

Class Interval	0 – 2	2 – 4	4 – 6	6 – 8	8 – 10
Frequency	2	3	5	3	2

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Options:

A.

$$\frac{463}{15}$$

B.

$$\frac{838}{15}$$

C.

$$\frac{44}{5}$$

D.

$$\frac{88}{15}$$

Answer: D

Solution:

Class interval	Frequency	x_i	$f_i x_i$
0 – 2	2	1	2
2 – 4	3	3	9
4 – 6	5	5	25
6 – 8	3	7	21
8 – 10	2	9	18
$\Sigma f_i = 15$			75

Mean (μ) = $\frac{75}{15} = 5$

$$\sigma^2 = \frac{\Sigma (x_i - \mu)^2 \cdot f_i}{N}$$
$$= \frac{32 + 12 + 0 + 12 + 32}{15} = \frac{88}{15}$$

Question7

The following data represents the frequency distribution of 20 observations

x_i	3	4	5	8	10	11
f_i	$\alpha + 2$	$(\alpha - 1)^2$	4	$\alpha - 1$	2	α

Then, its mean deviation about the mean is

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Options:

A.

3

B.

2.4

C.

2.7

D.

2.9

Answer: C

Solution:

We have, $\Sigma f = 20$

$$\therefore (\alpha + 2) + (\alpha - 1)^2 + 4 + (\alpha - 1) + 2 + \alpha = 20$$

$$\Rightarrow \alpha^2 + \alpha - 12 = 0$$

$$(\alpha + 4)(\alpha - 3) = 0 \Rightarrow \alpha = +3, -4$$

$$\therefore \alpha = 3 \quad [\because \alpha \neq -4]$$

x_i	f_i	$f_i x_i$	$ x_i - \bar{x} $	$f \cdot x_i - \bar{x} $
3	5	15	3	15
4	4	16	2	8
5	4	20	1	4
8	2	16	2	4
10	2	20	4	8
11	3	33	5	15
20	120		$\Sigma x_i - \bar{x} = 54$	

$$\bar{x} = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{120}{20} = 6$$

$$\text{MD} = \frac{\Sigma f_i |x_i - \bar{x}|}{\Sigma f} = \frac{54}{20} = 2.7$$

Question8

If the variance of the first n natural numbers is 10 and the variance of the first m even natural numbers is 16 , then $n : m =$

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Options:

A.

9 : 5

B.

7 : 3

C.

11 : 7

D.

5 : 8

Answer: C

Solution:

Variance of Ist n natural number is

$$= \frac{n^2-1}{12} = 10 \quad \dots (i)$$

and variance of Ist m even natural number is

$$\frac{m^2-1}{3} = 16 \quad \dots (iii)$$

On solving Eqs. (i) and (ii), we get

$$n^2 = 121 \text{ and } m^2 = 49$$

$$n : m = 11 : 7$$

Question9

The variance of ungrouped data 2, 12, 3, 11, 5, 10, 6, 7, is

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Options:

A.

11.875

B.

11

C.

12

D.

10.765

Answer: C

Solution:

Given, 2, 12, 3, 11, 5, 10, 6, 7

$$\text{Mean } \bar{x} = \frac{56}{8} = 7$$

$$\begin{aligned}\Sigma(x_i - \bar{x})^2 &= 25 + 25 + 16 + 16 + 4 + 9 + 1 + 0 \\ &= 96\end{aligned}$$

$$\text{var}(x) = \frac{(x_i - \bar{x})^2}{n} = \frac{96}{8} = 12$$

Question10

If the mean of the data 7, 8, 9, 7, 8, 7, λ and 8 is 8, then variance of the data is equal to

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Options:

A. 2

B. $7/8$

C. $9/8$

D. 1

Answer: D

Solution:

To find the variance of the data set 7, 8, 9, 7, 8, 7, λ , 8 with a mean of 8, follow these steps:

First, solve for λ :

$$\frac{7+8+9+7+8+7+\lambda+8}{8} = 8$$

Simplify the equation:

$$\frac{54+\lambda}{8} = 8$$

Multiply both sides by 8:

$$54 + \lambda = 64$$

Solve for λ :

$$\lambda = 64 - 54 = 10$$

Next, calculate the variance using the formula:

$$\text{Variance} = \frac{\sum x_i^2}{8} - \bar{x}^2$$

Where $\sum x_i^2$ is the sum of squares of each data point:

$$\begin{aligned}\sum x_i^2 &= 7^2 + 8^2 + 9^2 + 7^2 + 8^2 + 7^2 + 10^2 + 8^2 \\ &= 49 + 64 + 81 + 49 + 64 + 49 + 100 + 64 \\ &= 520\end{aligned}$$

Calculate the variance:

$$\frac{520}{8} - 8^2 = 65 - 64 = 1$$

Thus, the variance of the data set is 1.

Question11

Based on the following statements, choose the correct option.

Statement I The variance of the first n even natural numbers is $\frac{n^2-1}{4}$

.

Statement II The difference between the variance of the first 20 even natural numbers and their arithmetic mean is 112 .

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Options:

- A. Both Statements are true and II is a correct explanation of I
- B. Both Statements are true but II is not a correct explanation of I
- C. Statement I is true and Statement II is false
- D. Statement I is false and Statement II is true

Answer: D

Solution:

Sum of first n even natural number

$$= 2 + 4 + 6 + \dots$$

$$= 2(1 + 2 + 3 + \dots)$$

$$= 2 \times \frac{n(n+1)}{2} = n(n+1)$$

$$\therefore \text{Mean} = \frac{n(n+1)}{n} = (n+1)$$

$$\text{We know that variance} = \frac{1}{n}(\sum x_1)^2 - (\bar{x})^2$$

$$\begin{aligned}
&= \frac{1}{n}(2^2 + 4^2 + 6^2 + \dots) - (n+1)^2 \\
&= \frac{4}{n}(1^2 + 2^2 + 3^2 + \dots) - (n+1)^2 \\
&= \frac{4}{n} \times \frac{n(n+1)(2n+1)}{6} - (n+1)^2 \\
&= \frac{n+1}{3}[2(2n+1) - 2(n+1)] = \frac{n^2-1}{3}
\end{aligned}$$

$\therefore$ Difference between the variance of the first n even natural numbers any arithmetic mean
 $= \frac{399}{3} - 21 = 112$

So, statement I is false and II is true.

Question12

If $x_1, x_2, x_3, \dots, x_n$ are n observations such, that $\Sigma(x_i + 2)^2 = 28n$ and $\Sigma(x_1 - 2)^2 = 12n$, then the variance is

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Options:

A. 12

B. 14

C. 16

D. 20

Answer: A

Solution:

Given the equations:

$$\Sigma(x_i + 2)^2 = 28n$$

$$\Sigma(x_i - 2)^2 = 12n$$

Let's start by expanding and simplifying each equation.

First equation:

$$\begin{aligned}
\Sigma(x_i + 2)^2 &= \Sigma(x_i^2 + 4x_i + 4) \\
&= \Sigma x_i^2 + 4\Sigma x_i + 4n \\
&= 28n
\end{aligned}$$

Second equation:

$$\begin{aligned}\Sigma(x_i - 2)^2 &= \Sigma(x_i^2 - 4x_i + 4) \\ &= \Sigma x_i^2 - 4\Sigma x_i + 4n \\ &= 12n\end{aligned}$$

Let's introduce the following notation:

$$S = \Sigma x_i$$

$$S_2 = \Sigma x_i^2$$

Using these, we can set up the system of equations:

$$S_2 + 4S + 4n = 28n$$

$$S_2 - 4S + 4n = 12n$$

Subtract the second equation from the first:

$$\begin{aligned}(S_2 + 4S + 4n) - (S_2 - 4S + 4n) &= 28n - 12n \\ 8S &= 16n \\ S &= 2n\end{aligned}$$

Substituting $S = 2n$ back into the first equation:

$$\begin{aligned}S_2 + 4(2n) + 4n &= 28n \\ S_2 + 8n + 4n &= 28n \\ S_2 &= 28n - 12n \\ S_2 &= 16n\end{aligned}$$

Now, to find the variance σ^2 , use the formula:

$$\begin{aligned}\sigma^2 &= \frac{1}{n} \Sigma (x_i - \mu)^2 \\ &= \frac{1}{n} \Sigma x_i^2 - \mu^2\end{aligned}$$

Since $\Sigma x_i = S = 2n$, the mean μ is:

$$\mu = \frac{\Sigma x_i}{n} = \frac{2n}{n} = 2$$

Therefore, the variance is:

$$\begin{aligned}\sigma^2 &= \frac{1}{n} \times 16n - 2^2 \\ &= 16 - 4 \\ &= 12\end{aligned}$$

Question13

The coefficient of variation for the frequency distribution is

x_i	4	3	1
f_i	1	3	5

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Options:

A. $\frac{50}{\sqrt{3}}$

B. $\frac{125}{2\sqrt{3}}$

C. $\frac{100}{3\sqrt{2}}$

D. $\frac{100}{\sqrt{3}}$

Answer: D

Solution:

Given frequency distribution

x_i	4	3	1
f_L	1	3	5

x_i	f_i	$f_i x_i$	$(x_i - \bar{x})$	$(x_i - \bar{x})^2$	$f_i(x_i - \bar{x})^2$
4	1	4	2	4	4
3	3	9	1	1	3
1	5	5	-1	1	5
	$\Sigma_i = 9$	$\Sigma \Sigma_i x_i = 18$			12

$\therefore$ Mean $\bar{x} = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{18}{9} = 2$

Now, standard deviation

$\sigma = \sqrt{\frac{\Sigma f_i (x_i - \bar{x})^2}{\Sigma f_i}} = \sqrt{\frac{12}{9}} = \frac{2}{\sqrt{3}}$

So, coefficient of variance = $\frac{\sigma}{\bar{x}} \times 100$

= $\frac{2}{\sqrt{3}} \times \frac{1}{2} \times 100 = \frac{100}{\sqrt{3}}$

Question14

Mean deviation about the mean for the following data is

Class Interval	0 – 6	6 – 12	12 – 18	18 – 24	24 – 30
Frequency	1	2	3	2	1

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Options:

A. 5

B. $16/3$

C. 6

D. $19/3$

Answer: B

Solution:

Mean $A = 15, h = 6$

Class Interval	Mid value x_i	Frequency	$d_i = \frac{x_i - 15}{6}$	fd_i	$x_i - \bar{x}$	$\frac{f_i}{x_i - \bar{x}}$
0 – 6	3	1	-2	-2	12	12
6 – 12	9	2	-1	-2	6	12
12 – 18	15	3	0	0	0	0
18 – 24	21	2	1	2	6	12
24 – 30	27	1	2	2	12	12
		$N = 9$		0	36	48

Mean, $\bar{X} = A + h \left(\frac{1}{N} \sum f_i d_i \right) = 15 + 6 \left[\frac{1}{9} (0) \right]$

$\bar{X} = 15$

Mean-deviation about mean

$$\begin{aligned} &= \frac{1}{N} \sum f_i |x_i - \bar{X}| = \frac{1}{9} \times 48 \\ &= \frac{1}{9} \times 48 = \frac{16}{3} \end{aligned}$$

Question15

If the mean deviation about the mean is m and variance is σ^2 for the following data, then $m + \sigma^2 =$

x	1	3	5	7	9
f	4	24	28	16	8

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Options:

A. 8

B. 7.2

C. $\frac{28}{5}$

D. 6

Answer: D

Solution:

Given data is

x	1	3	5	7	9
f	4	24	28	16	8

$$\text{Now, } \bar{x} = \frac{4+72+140+112+72}{80}$$

$$= \frac{400}{80} = 5$$

$$\text{Thus, mean deviation about mean is } m = \frac{4 \times 4 + 24 \times 2 + 28 \times 0 + 16 \times 2 + 8 \times 4}{80} = \frac{8}{5}$$

$$\text{and } \sigma^2 = \frac{4 \times 4^2 + 24 \times 2^2 + 28 \times 0^2 + 16 \times (-2)^2 + 8 \times (-4)^2}{80} = \frac{22}{5}$$

$$\text{Now, } m + \sigma^2 = \frac{8}{5} + \frac{22}{5} = \frac{30}{5} = 6$$

Question16

$\bar{x}$ and $\bar{y}$ are the arithmetic means of the runs of two batsmen A and B in 10 innings respectively and σ_A, σ_B are the standard deviations of their runs in them. If batsman A is more consistent than B , then he is also a higher run scorer only when

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Options:

A. $0 < \frac{\sigma_A}{\sigma_B} < \frac{\bar{x}}{\bar{y}}, \frac{\bar{x}}{\bar{y}} > 1$

B. $\frac{\bar{x}}{\bar{y}} > \frac{\sigma_A}{\sigma_B} > 1$

C. $\frac{\bar{x}}{\bar{y}} < \frac{\sigma_A}{\sigma_B} < 1$

D. $\frac{\bar{x}}{\bar{y}} > 1, 1 \leq \frac{\sigma_A}{\sigma_B} < \frac{\bar{x}}{\bar{y}}$

Answer: A

Solution:

Given that $\bar{x}$ and $\bar{y}$ are the arithmetic means of the runs for batsmen A and B , respectively, and σ_A and σ_B represent the standard deviations of their runs.

Since Batsman A is more consistent than Batsman B , we have:

$$\sigma_A < \sigma_B \Rightarrow \frac{\sigma_A}{\sigma_B} < 1$$

Additionally, we know that if Batsman A is a higher run scorer than Batsman B , then:

$$\bar{x} > \bar{y} \Rightarrow \frac{\bar{x}}{\bar{y}} > 1$$

Combining both conditions, we get:

$$0 < \frac{\sigma_A}{\sigma_B} < \frac{\bar{x}}{\bar{y}} \quad \text{and} \quad \frac{\bar{x}}{\bar{y}} > 1$$

This indicates that Batsman A scores more runs only when the ratio of the standard deviations is less than the ratio of the arithmetic means, with the latter being greater than 1.

Question17

If m and M denote the mean deviations about mean and about median respectively of the data 20, 5, 15, 2, 7, 3, 11, then the mean deviation about the mean of m and M is

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Options:

A. $\frac{1}{7}$

B. $\frac{38}{7}$

C. $\frac{36}{7}$

D. $\frac{37}{7}$

Answer: A

Solution:

We have, 20, 5, 15, 2, 7, 3, 11.

$$\begin{aligned}\text{Mean} &= \frac{20 + 5 + 15 + 2 + 7 + 3 + 11}{7} \\ &= \frac{63}{7} = 9\end{aligned}$$

Data in ascending order = 2, 3, 5, 7, 11,

15, 20

Median = 7

m = Mean deviations about means

$$\begin{aligned}&|20 - 9| + |5 - 9| + |15 - 9| + |2 - 9| \\ &+ |7 - 9| + |3 - 9| + |11 - 9| \\ &= \frac{11 + 4 + 6 + 7 + 2 + 6 + 2}{7} = \frac{38}{7}\end{aligned}$$

M = Mean deviation about median

$$\begin{aligned}M &= \frac{|20 - 7| + |5 - 7| + |15 - 7| + |2 - 7|}{7} \\ &+ \frac{|7 - 7| + |3 - 7| + |11 - 7|}{7} \\ &= \frac{13 + 2 + 8 + 5 + 0 + 4 + 4}{7} = \frac{36}{7}\end{aligned}$$

$$\text{Mean of } m \text{ and } M = \left(\frac{38}{7} + \frac{36}{7}\right) \frac{1}{2} = \frac{37}{7}$$

MD (Mean of m and M)

$$\begin{aligned}
&= \frac{1}{2} \left\{ \left| \frac{36}{7} - \frac{37}{7} \right| + \left| \frac{38}{7} - \frac{37}{7} \right| \right\} \\
&= \frac{1}{2} \times \frac{2}{7} = \frac{1}{7}
\end{aligned}$$

Question18

For a set of observations, if the coefficient of variation is 25 and mean is 44 , then the variance is

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Options:

- A. 11
- B. 121
- C. 110
- D. 19

Answer: B

Solution:

To find the variance given the coefficient of variation (CV) and the mean, we can use the following relationship:

The formula for the coefficient of variation (CV) is given by:

$$CV = \frac{\sigma}{\mu} \times 100$$

where σ is the standard deviation and μ is the mean.

In this problem, the coefficient of variation is 25, and the mean (μ) is 44. Substituting these values into the equation, we have:

$$\frac{\sigma}{44} \times 100 = 25$$

Solving for σ , we get:

$$\sigma = \frac{25 \times 44}{100} = 11$$

The variance (σ^2) is the square of the standard deviation:

$$\sigma^2 = 11^2 = 121$$

Therefore, the variance is 121.

Question19

If the mean deviation of the data

$1, 1 + d, 1 + 2d, \dots, 1 + 100d, (d > 0)$ from their mean is 255, then 'd' is equal to

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Options:

A. 10.1

B. 10.2

C. 10.3

D. 10.4

Answer: A

Solution:

$$\text{Mean } \bar{x} = \frac{\text{Sum of quantities}}{n}$$

$$= \frac{\frac{n}{2}(a+l)}{n} \quad [l = \text{last term}]$$

$$= \frac{1}{2}(a + l) = \frac{1}{2}(1 + 1 + 100d)$$

$$= 1 + 50d$$

$$\text{MD} = \frac{1}{n} \sum |x_i - \bar{x}|$$

$$\Rightarrow 255 = \frac{1}{101} [50d + 49d + 48d + \dots + d + 0 + d + \dots + 50d]$$

$$\Rightarrow 255 = \frac{2d}{101} \left[\frac{50 \times 51}{2} \right]$$

$$d = \frac{255 \times 101}{50 \times 51} = 10.1$$

Question20

If the mean of the data $p, 6, 6, 7, 8, 11, 15, 16$, is 3 times p , then the mean deviation of the data from its mean is

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Options:

A. 2.25

B. 3.75

C. 4.4

D. 2.5

Answer: B

Solution:

Given, data of number are $p, 6, 6, 7, 8, 11, 15$ and 16 .

$$\begin{aligned}\text{Mean } \bar{x} &= \frac{\text{Sum of all observations}}{\text{Total number of observations}} \\ &= \frac{p + 6 + 6 + 7 + 8 + 11 + 15 + 16}{8}\end{aligned}$$

$$\Rightarrow \frac{69 + p}{8} = 3p \quad (\text{given})$$

$$\Rightarrow 24p - p = 69 \Rightarrow 23p = 69$$

$$\Rightarrow p = 3$$

So given data is $3, 6, 6, 7, 8, 11, 15, 16$ and mean is 9 .

So, the mean deviation

$$\begin{aligned}& \frac{|3 - 9| + |6 - 9| + |6 - 9| + |7 - 9| + |8 - 9|}{8} \\ & + \frac{|11 - 9| + |15 - 9| + |16 - 9|}{8} \\ &= \frac{6 + 3 + 3 + 2 + 1 + 2 + 6 + 7}{8} = \frac{30}{8} = 3.75\end{aligned}$$

Question21

The mean deviation about the mean for the following data.

5, 6, 7, 8, 6, 9, 13, 12, 15 is

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Options:

A. 1.55

B. 2.88

C. 3.89

D. 5

Answer: B

Solution:

Mean,

$$\begin{aligned}\bar{x} &= \frac{5 + 6 + 7 + 8 + 6 + 9 + 13 + 12 + 15}{9} \\ &= \frac{81}{9} = 9\end{aligned}$$

Deviation, $x_i - \bar{x}$:

$$\begin{aligned}5 - 9, 6 - 9, 7 - 9, 8 - 9, 6 - 9, 9 - 9, 13 - 9, 12 - 9, 15 - 9 \\ \Rightarrow -4, -3, -2, -1, -3, 0, 4, 3, 6 \\ |x_i - \bar{x}| : 4, 3, 2, 1, 3, 0, 4, 3, 6\end{aligned}$$

$$\begin{aligned}\text{MD} &= \frac{\sum_{i=1}^9 |x_i - \bar{x}|}{9} \\ &= \frac{4 + 3 + 2 + 1 + 3 + 0 + 4 + 3 + 6}{9} \\ &= \frac{26}{9} = 2.88\end{aligned}$$

Question22

If the mean of a data x is 10 and if all the observations are multiplied by 2, then the mean of new data is

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Options:

A. 30

B. 15

C. 50

D. 20

Answer: D

Solution:

Let data be

$$x_1, x_2, x_3, \dots, x_n$$

Given that,

$$\frac{x_1 + x_2 + x_3 + \dots + x_n}{n} = 10$$
$$\Rightarrow x_1 + \dots + x_n = 10n$$

When all observation multiplied by 2, we obtain new data as $2x_1, 2x_2, \dots, 2x_n$ then

$$\begin{aligned} \text{Mean} &= \frac{2x_1 + 2x_2 + \dots + 2x_n}{n} \\ &= \frac{2(x_1 + \dots + x_n)}{n} = \frac{2(10n)}{n} = 20 \end{aligned}$$

Question23

The mean deviation from the mean of the set of observation $-1, 0, 4$ is

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Options:

A. 3

B. 1

C. -2

D. 2

Answer: D

Solution:

$$\text{Mean} = \frac{-1+0+4}{3} = 1$$

$$\text{Deviation} = |-1 - 1|, |0 - 1|, |4 - 1| = 2, 1, 3$$

$$\text{Mean deviation} = \frac{2+1+3}{3} = 2$$

Question24

Let an angle of a triangle is 60° . If the variance of the angles of the triangle is 1014° , then the other two angles are

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Options:

A. 23° and 97°

B. 22° and 68°

C. 21° and 99°

D. 20° and 100°

Answer: C

Solution:

$$\text{Variance} = \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n} \right)^2$$

$$1014 = \frac{60^2 + a^2 + b^2}{3} - 60^2$$

$$10242 = a^2 + b^2 \dots (i)$$

$$a + b = 120 \dots (ii)$$

From Eqs. (i) and (ii), we get

$$a = 21 \text{ and } b = 99$$

Question 25

For the random variable X with probability distribution is given by the table

$X = x$	0	1	2	3
$P(X = x)$	K	$K + \frac{1}{7}$	$2K$	$\frac{2}{5}$

The mean of X is

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Options:

A. $\frac{31}{35}$

B. $\frac{57}{35}$

C. $\frac{63}{35}$

D. $\frac{67}{35}$

Answer: D

Solution:

$$\begin{aligned}
 \Sigma P(X = r) &= 1 \\
 K + K + \frac{1}{7} + 2K + \frac{2}{5} &= 1 \Rightarrow 4K = \frac{3}{5} - \frac{1}{7} \\
 4K &= \frac{16}{35} \Rightarrow K = \frac{4}{35} \\
 \alpha &= \frac{\Sigma P_i X_i}{\Sigma P_i} = \Sigma P_i X_i = K + \frac{1}{7} + 4K + \frac{6}{5} \\
 &= 5K + \frac{47}{35} = 5 \left(\frac{4}{35} \right) + \frac{47}{35} = \frac{67}{35}
 \end{aligned}$$

Question 26

The mean and variance of n observations $x_1, x_2, x_3, \dots, x_n$ are 5 and 0 respectively. If $\sum_{i=1}^n x_i^2 = 400$, then the value of n is equal to

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Options:

A. 80

B. 25

C. 20

D. 16

Answer: D

Solution:

Mean $\bar{x} = 5$ and $\Sigma x^2 = 400$

$$\therefore \text{var}(x) = 0$$

$$\therefore \text{var}(x) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$0 = \frac{1}{n} \left[\sum_{i=1}^n x_i^2 + \sum_{i=1}^n \bar{x}^2 - 2\bar{x} \sum_{i=1}^n x_i \right]$$

$$0 = \frac{1}{n} [400 + 25n - 2 \times 5 \times n\bar{x}]$$

$$0 = \frac{1}{n} [400 + 25n - 50n]$$

$$\Rightarrow \frac{400}{n} = 25$$

$$\Rightarrow n = 16$$

Question27

The variance of the variates 112, 116, 120, 125 and 132 about their AM is

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Options:

A. 58.8

B. 60

C. 48.8

D. 61.8

Answer: C

Solution:

x	$(x - \bar{x})^2$
112	81
116	25
120	1
125	16
132	121
$\Sigma x = 605$	$\Sigma(x - \bar{x})^2 = 244$

$$\text{Mean} = \bar{x} = \frac{605}{5} = 121$$

$$\text{Variance} = \frac{\Sigma(x - \bar{x})^2}{n} = \frac{244}{5} = 48.8$$

Question28

Which of the following set of data has least standard deviation?

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Options:

A. 10, 20, 30, 40

B. 2, 4, 6, 8

C. 3, 6, 9, 12

D. 1, 2, 3, 4

Answer: D

Solution:

Clearly, difference of values from arithmetic mean is least in option (d).

⇒ Option (d) has least standard deviation.
